HR Analytics & Employee Attrition Analysis

Python Data Analyst Portfolio Project

1. Project Overview

This project analyzes employee HR data to identify patterns associated with employee attrition. The analysis covers attrition, department, job role, salary, job satisfaction, performance, age, experience, correlations, and statistical tests.

2. Business Objectives

• Measure the overall employee attrition rate.

• Identify departments and job roles with higher attrition.

• Compare salary and job satisfaction between employees who stayed and left.

• Analyze age and experience patterns.

• Use statistical testing to evaluate selected relationships.

3. Dataset Summary

Metric	Value
Rows	4,327
Columns	31
Employees Stayed	3,626
Employees Left	701
Attrition Rate	16.20%
Average Age	36.92
Average Monthly Income	65,029.24
Average Years at Company	7.02
Average Job Satisfaction	2.72
Average Performance Rating	3.15

4. Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, and SciPy.

5. Key Findings

• The overall attrition rate is 16.20% (701 employees left).

• Human Resources has the highest department-level attrition rate at 29.79%.

• Research Director has the highest job-role attrition rate at 23.95%.

• Average monthly income is 65,724.66 for employees who stayed and 61,432.14 for employees who left.

• Average job satisfaction is 2.77 for employees who stayed and 2.47 for employees who left.

• Average age is 37.55 for employees who stayed and 33.66 for employees who left.

• Average years at company is 7.38 for employees who stayed and 5.14 for employees who left.

6. Statistical Analysis

Test	Statistic	P-value	Interpretation
Monthly Income Welch t-test	2.307	0.021228	Significant
Job Satisfaction Welch t-test	6.727	2.94053e-11	Significant
Department vs Attrition Chi-square	27.117	1.29279e-06	Significant association

The statistical tests show significant differences in monthly income and job satisfaction between employees who stayed and left. The chi-square test also indicates a statistically significant association between department and attrition. These results indicate association, not causation.

7. Visualizations

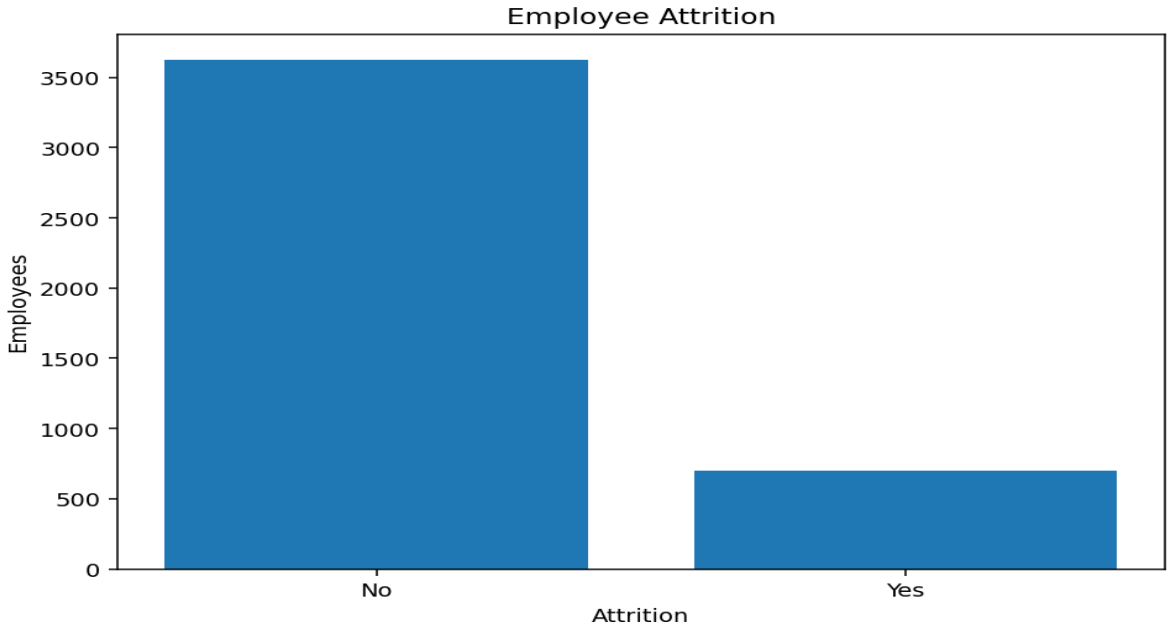


Figure 1. Employee Attrition

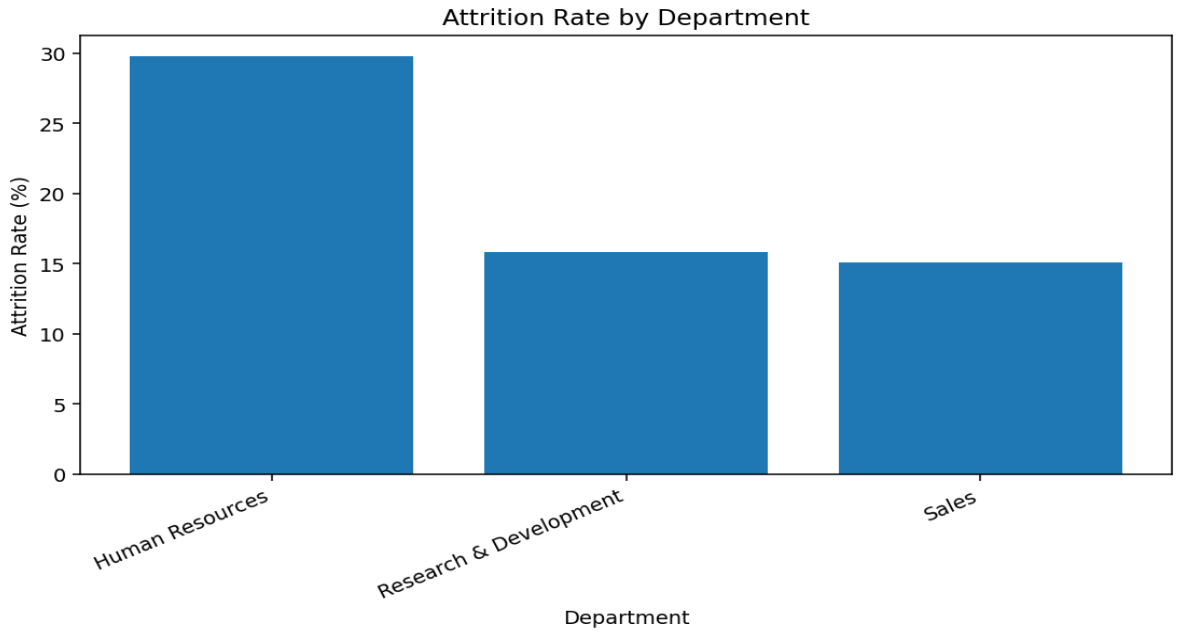


Figure 2. Attrition Rate by Department

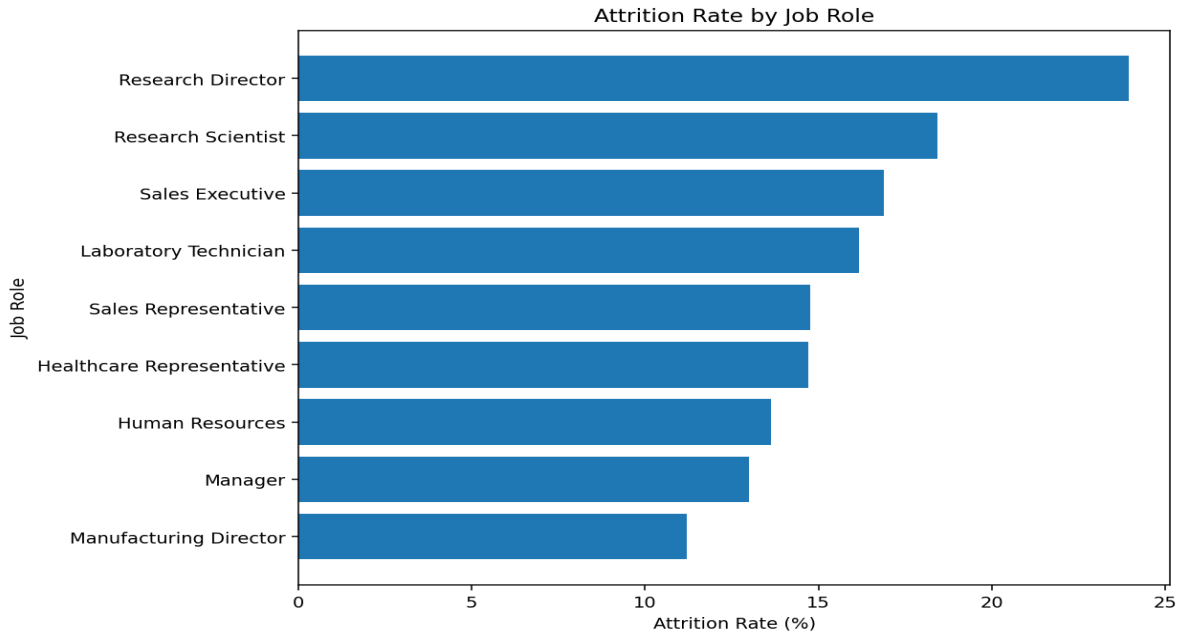


Figure 3. Attrition Rate by Job Role

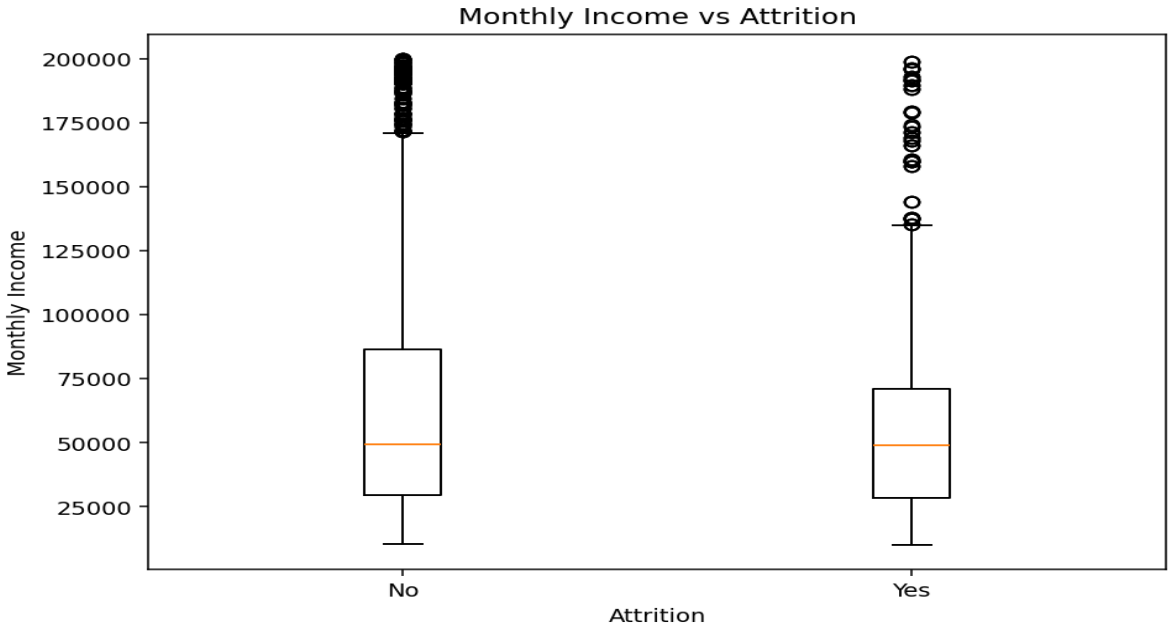


Figure 4. Monthly Income vs Attrition

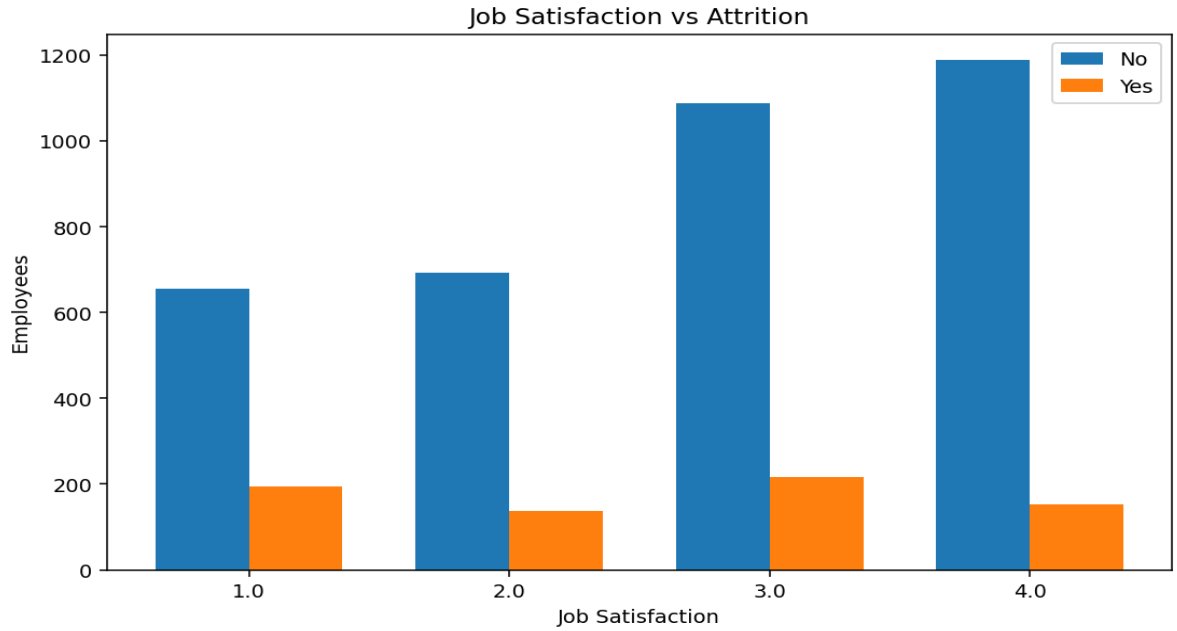


Figure 5. Job Satisfaction vs Attrition

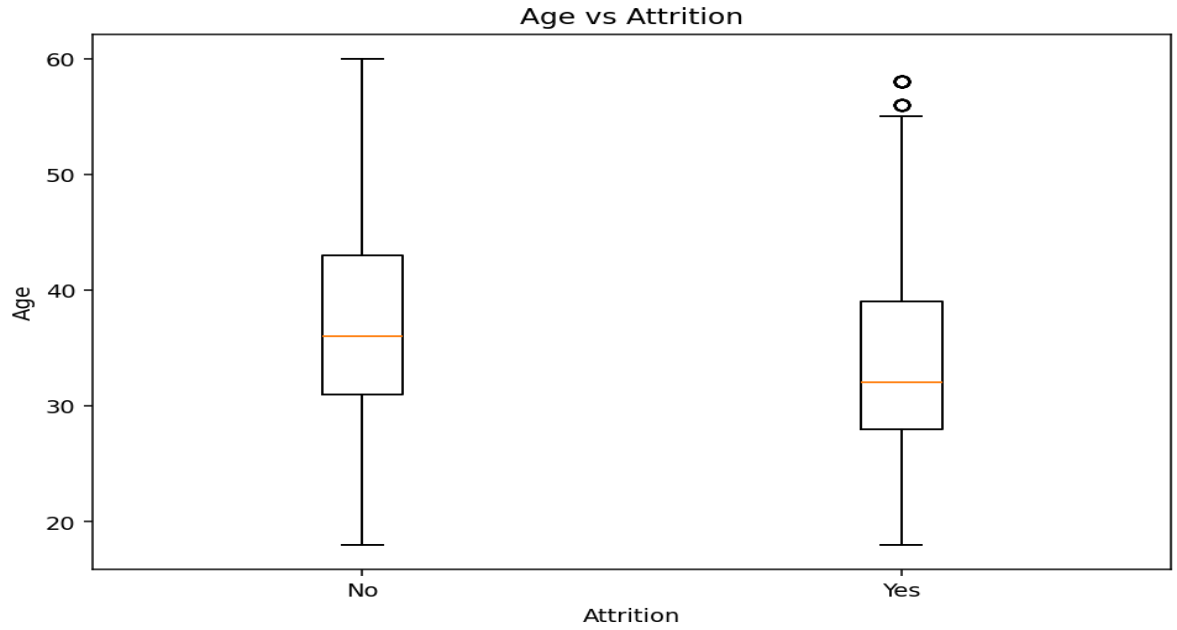


Figure 6. Age vs Attrition

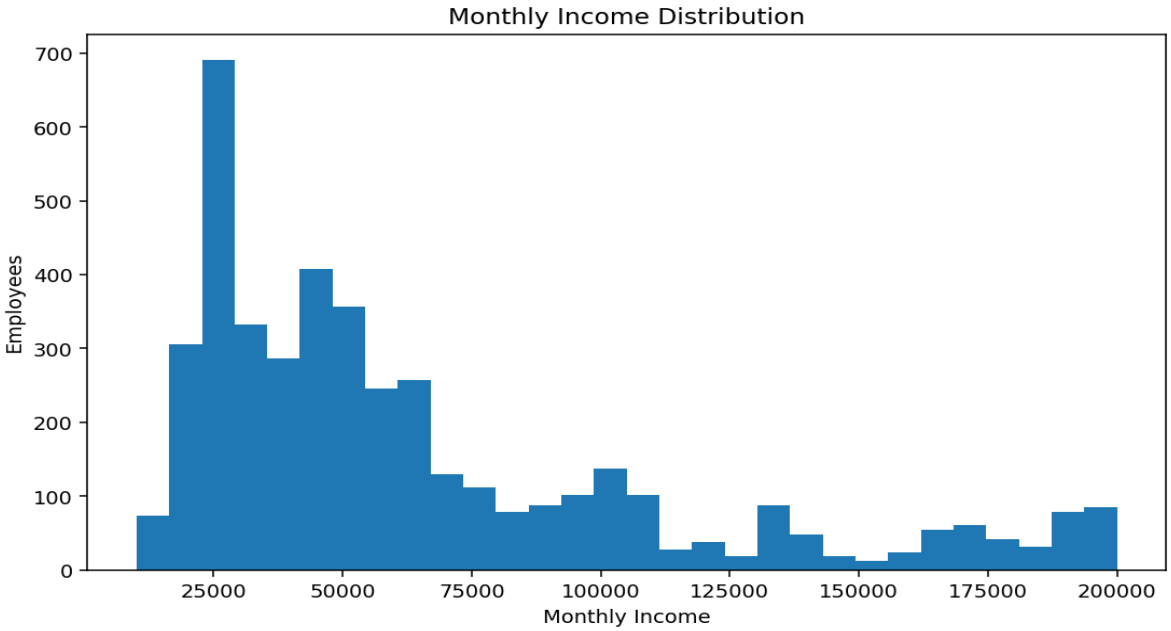


Figure 7. Monthly Income Distribution

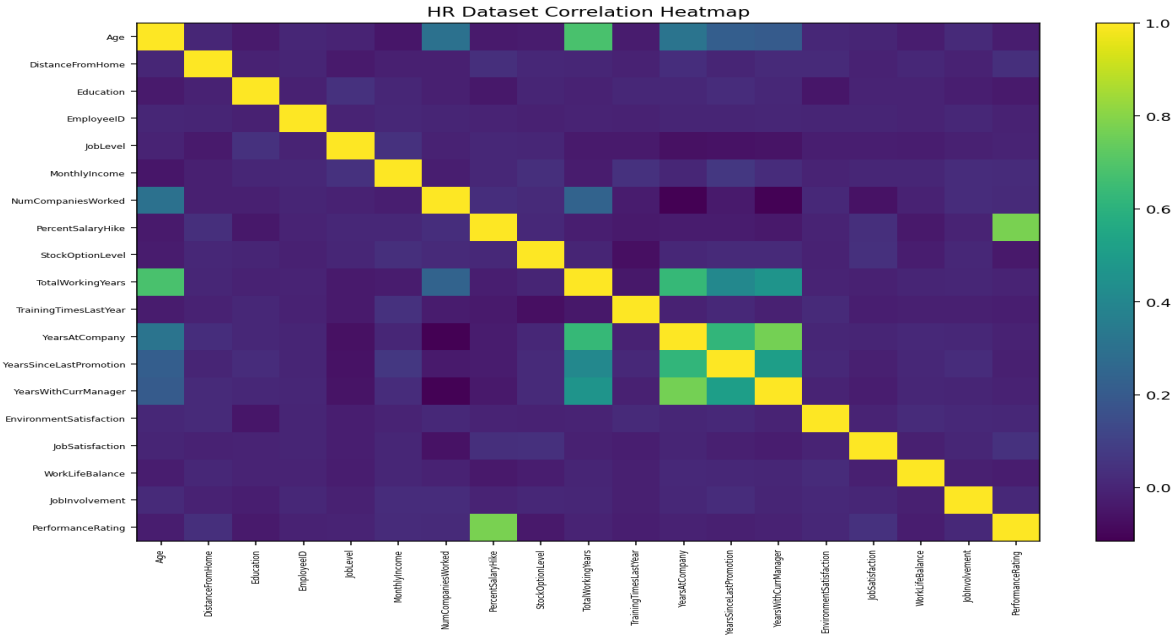


Figure 8. HR Dataset Correlation Heatmap

8. Business Recommendations

• Prioritize retention analysis in Human Resources, which has the highest department-level attrition rate in this dataset.

• Investigate retention factors affecting Research Director, which has the highest job-role attrition rate.

• Review compensation patterns for groups with higher attrition, because monthly income differs significantly between the two attrition groups.

• Monitor job satisfaction and investigate workplace factors linked to lower satisfaction.

• Pay attention to early-tenure employees, since employees who left have lower average years at the company.

• Use these findings as decision-support evidence and investigate additional factors before making causal claims.

9. Conclusion

The project demonstrates an end-to-end Python Data Analyst workflow: dataset understanding, descriptive analysis, attrition analysis, salary and satisfaction analysis, visualization, correlation analysis, hypothesis testing, and business recommendations. The results can support HR teams in identifying areas for deeper retention analysis.